STAT-S 670: EXPLORATORY DATA ANALYSIS

FINAL-PROJECT: CUSTOMER SEGMENTATION ANALYSIS

Abstract

This project investigates the spending behaviors of customers relative to their income, age, and marital status using the Customer Personality Dataset from Kaggle. Through a detailed exploratory analysis, we identified significant patterns in spending relative to demographic factors and interaction with marketing efforts. We employed various statistical and machine learning techniques, including Principal Component Analysis (PCA), Generalized Additive Models (GAM), and multiple regression models, to uncover the nuanced relationships between customer demographics and their spending behaviors. Our findings suggest that age and income are significant predictors of spending, with variances in spending behaviors observable across different marital statuses. These insights can assist businesses in tailoring their marketing strategies to better align with the financial capabilities and preferences of distinct customer segments. The methodologies and results detailed in this report not only contribute to academic knowledge but also offer practical strategies for effective customer segmentation and targeted marketing.

Objective

Understanding customer spending relative to income and demographic variables like age and marital status helps businesses tailor their marketing strategies more effectively. This segmentation leads to targeted advertising, product placement, and pricing strategies that align better with customer needs and financial capabilities.

Research Questions

1. Spending vs. Income Categorized by Age
How does spending behavior vary across different age groups with varying household incomes? Is there a linear relationship between spending and income for each age group, or is a more complex model required to understand the nuances? Are there unique spending behaviors specific to certain age groups? If so, what are these behaviors and what could be driving them?

2. Spending vs. Income Categorized by Marital Status
How does marital status influence spending relative to income? Is the spending-income relationship for different marital statuses best explained by a straightforward linear model, or does it require a more complex model to capture nuances? Are there identifiable patterns in spending behaviors among different marital statuses, and what might explain any variations?

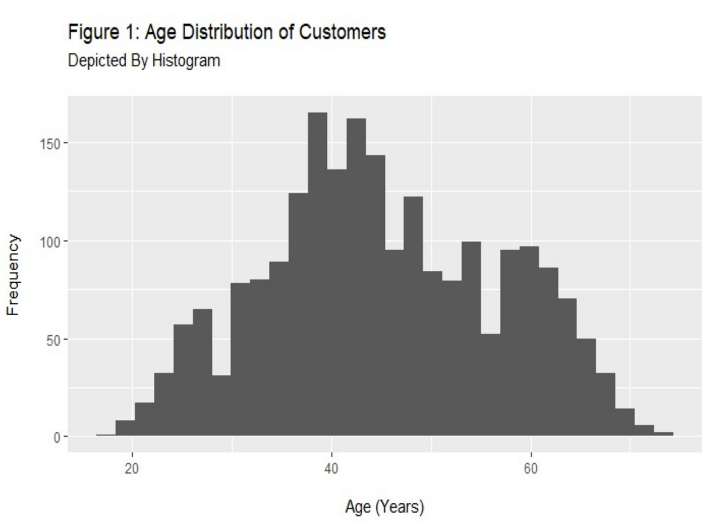
3. Customer Segmentation
What distinct segments of customers can be identified based on their purchasing behaviors, demographics, and interactions with marketing campaigns? What unique characteristics define each identified customer segment?

Dataset Description

"Customer Personality Dataset" on Kaggle is provided by Dr. Omer Romero-Hernandez. This dataset has 29 variables where 25 are quantitative and it has 2240 observations. The dataset includes a survey of customers of a business, where the purchase and spending information is represented in US dollars. This dataset serves as a valuable resource for understanding and targeting diverse customer groups based on their characteristics and behaviors.

Key Variables Taken Into Consideration for Analysis

- **Age:** Customer's Age (Years)
- **Income:** Customer's Yearly Household Income (US dollars)
- **Marital Status:** Customer's Relationship Status (Single, Widow, Divorced, Couple)
- **Amount Spent:** Amount Spent by the Customer on Different Products (Fruits, Wines, Meat, Fish, Sweets, Gold) (US dollars)
- **Customer's Channels of Purchase:** Web, Store, Deals and Catalog



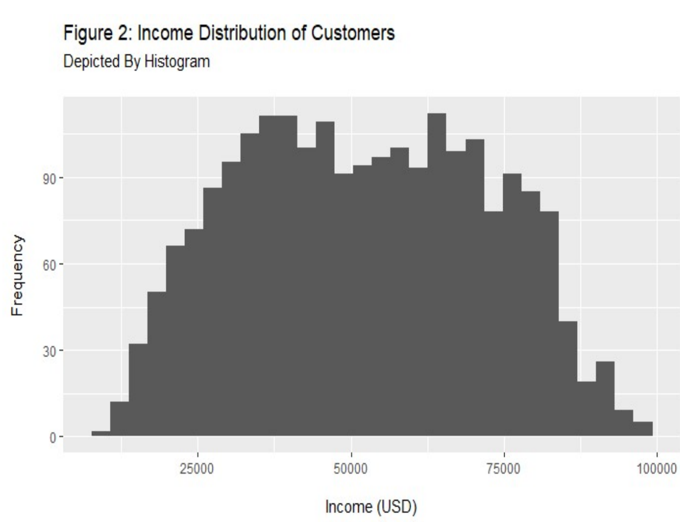


Figure 1: Age Distribution of Customers - The histogram illustrates the age distribution within our customer database, with the most significant density centering around the 40-year mark, suggesting that our primary customer segment falls within the middle-aged demographic, particularly those between 30 to 50 years of age. The representation decreases for the younger (<25 years) and older (>60 years) populations. This pattern emphasizes the potential for tailoring marketing strategies to the predominant age groups, enhancing engagement and possibly boosting transactions with targeted offerings.

Figure 2: Income Distribution of Customers - This histogram provides a visual representation of our customers' income levels, exhibiting a bimodal distribution with prominent peaks around the $37,000 and $62,000 marks. This pattern highlights a customer base primarily concentrated in these middle-income brackets. The steep decline post the second peak reflects a significant drop in customers within higher income tiers, particularly above $75,000. The long right tail of the distribution suggests a smaller yet notable segment of customers with higher incomes. Recognizing these income segments is vital for crafting marketing strategies and pricing models that cater effectively to the predominant income groups in our customer base.

Figure 3: Distribution of Customers Based on Their Marital Status - The pie chart delineates our customer base's marital status composition, with the largest segment, 64.7%, represented in light blue, encompassing individuals in a partnership or marriage i.e. Couple. Singles, visualized in dark blue, make up the second-largest group at 21.4%, indicating a significant single customer demographic. Those who are divorced, shown in orange, form a notable 10.4% of the customer base, while widows, indicated in green, comprise the smallest proportion at 3.5%. This distribution is key for tailoring marketing strategies to effectively address the different lifestyle needs and preferences associated with each marital status group.

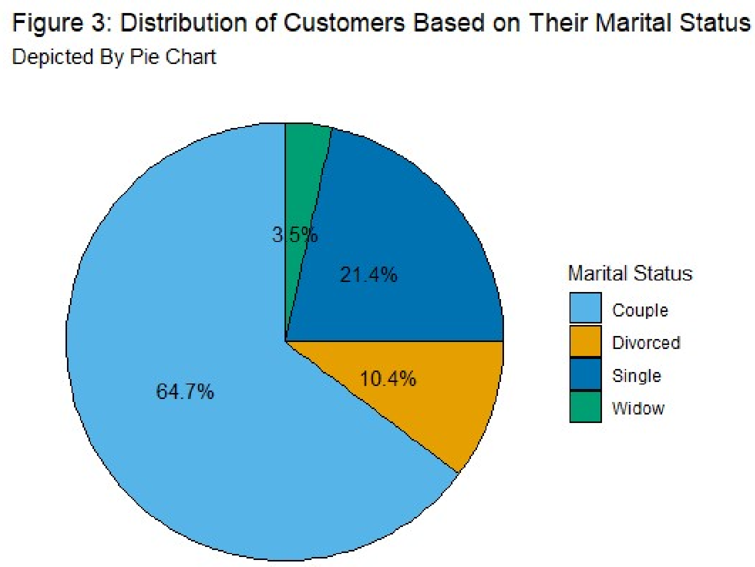
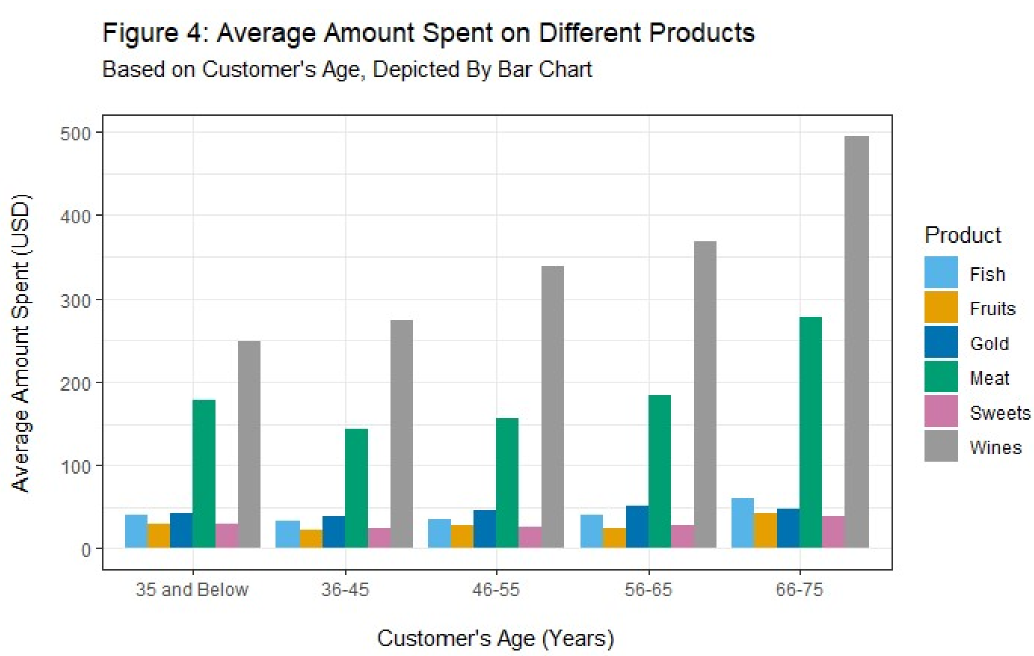


Figure 4: Average Amount Spent on Different Product, Based on Customer's Age - The chart shows the average amount spent in USD on six different products, categorized by the age groups of the customers.



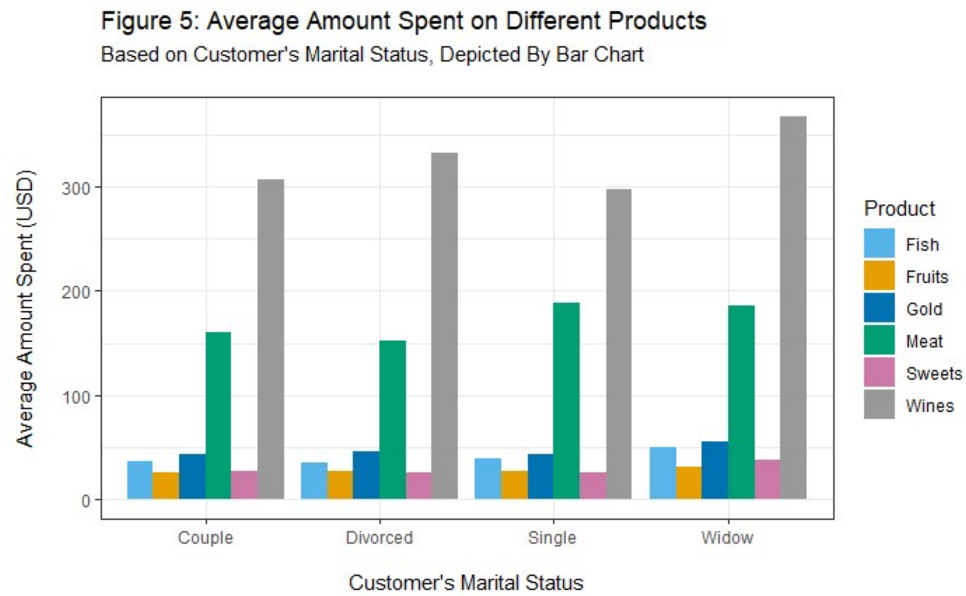
Key Observations:

- The youngest age group, **35 and Below,** spends the least on average, with very low expenditures on gold and wines. Whereas the **66-75** age group is the highest spender, with notably high expenditure on wines.
- The **46-55** age groups spends the most on average across all products, particularly on gold, which shows a significant spike compared to other age groups.
- For the **56-65** age groups, meat and wines appear to be the most purchased products.

Spending on Wine emerges as the top choice and Meat is a second most choice across all demographics. Spending on fruits is relatively consistent across all age groups but is not the highest expenditure in any category. Sweets have the least variation in spending across different age groups.

This bar chart is useful for understanding consumer spending habits across different age brackets and could be used by marketers to tailor their strategies according to customer demographics.

Figure 5: Average Amount Spent on Different Product, Based on Customer's Marital Status - This chart illustrates the average amount of money in USD spent on various products, segmented by the marital status of the customers.



Observations from the chart include:

- Customers who are **widowed** spend the most on average, particularly on wines and gold, which is significantly higher than any other category.
- **Divorced** customers have the second-highest spending on gold and also show substantial spending on wines.
- **Couples** and **single** customers spend less on average compared to divorced and widowed customers.
- Spending on fish, fruits, and sweets appears relatively moderate across all marital statuses.
- **Single** customers have the lowest average spending in most categories, especially on gold and wines.

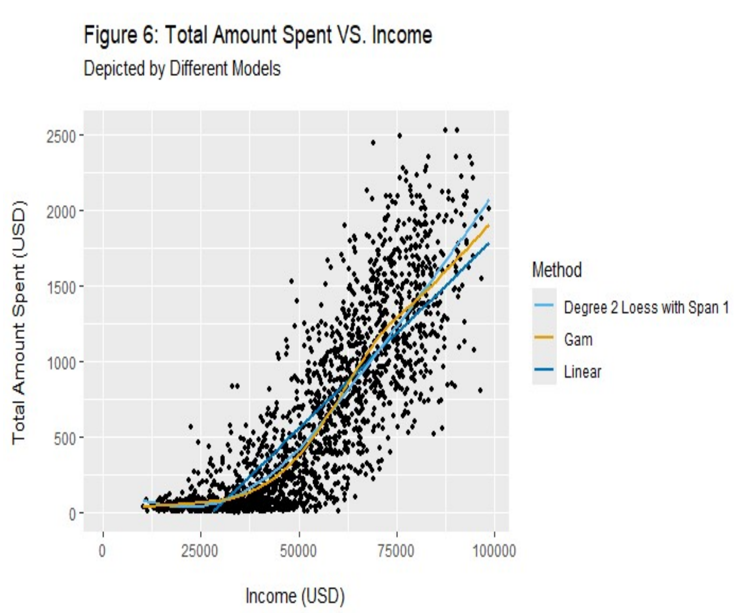
The expenditure pattern for meat is quite uniform across all groups, with singles spending slightly more than others. Additionally, wines are particularly popular, dominating expenditures across all groups.

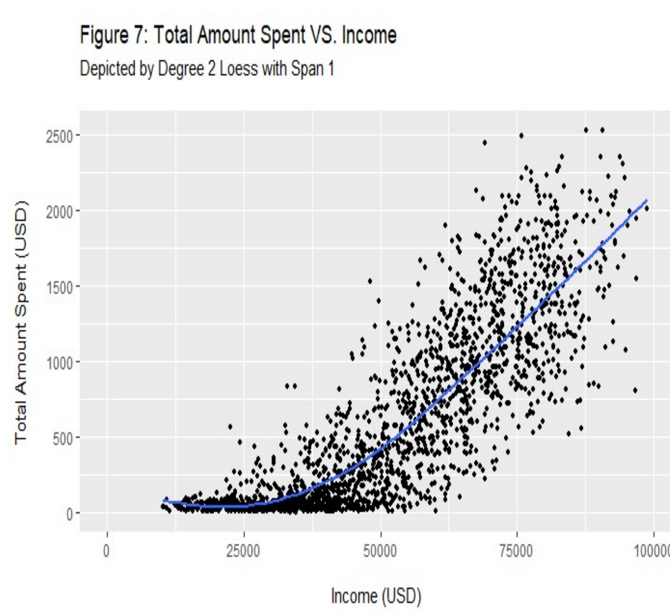
The chart can provide valuable insights for businesses seeking to understand spending patterns based on marital status, allowing them to tailor their products and marketing strategies to different customer segments.

Research Question 1

- **How does spending behavior vary across different age groups with varying household incomes?**
- **Is there a linear relationship between spending and income for each age group, or is a more complex model required to understand the nuances?**

Relationship between Total Amount Spent and Income





The depicted plot (Figure 6) illustrates the relationship between Total Amount Spent and Income, analyzed using three different models. The linear model is represented by the dark blue line, a loess curve of degree 2 with a span of 1 is depicted in light blue, and the orange curve represents a gam curve.

Upon careful examination of the data point distribution, it becomes apparent that the linear model is unsuitable choices, as they fail to capture the evident non-linear trend in the data.

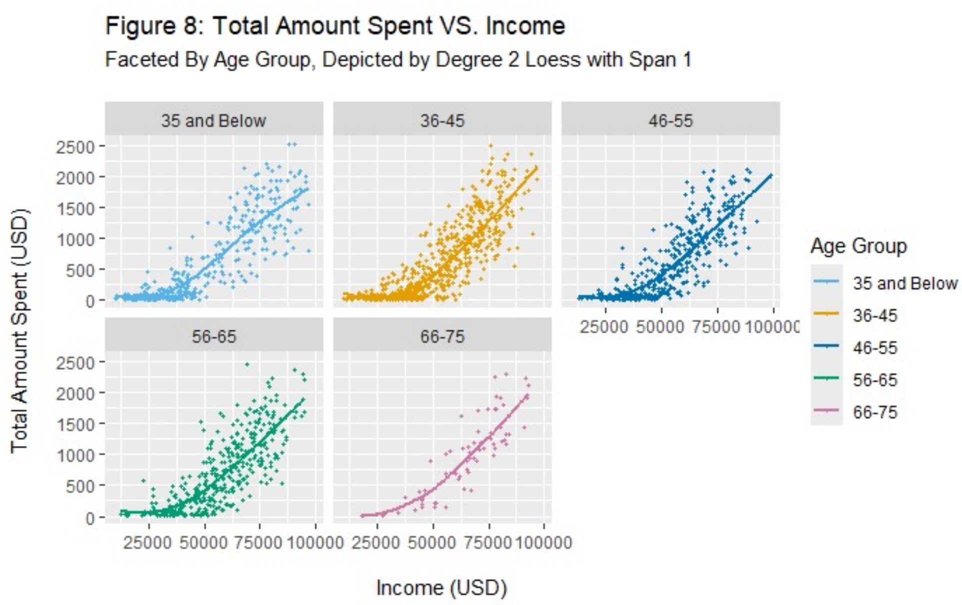
The scatter plot, featuring data points along with various fit lines, reveals that the light blue curve (loess with a span of 1 and degree 2) strikes a good balance. It adeptly captures the overall trend while accommodating local variations. In contrast, the orange curve (gam) is more susceptible to overfitting noise.

Therefore, we observe that the **degree 2 loess model (i.e. Curved Function) with a span of 1** is good enough to define the relationship between Total Amount Spent and Income.

The scatter plot (Figure 7) and accompanying **Degree 2 Loess curve with a span of 1** provide a nuanced visualization of the relationship between income and total amount spent by customers. The curve indicates a progressive, non-linear increase in spending as income grows. The gentle slope at lower income levels suggests a more conservative spending pattern, which accelerates noticeably as income rises, especially beyond the $50,000 mark. This inflection point may reflect a threshold where disposable income sufficiently increases, thus allowing for greater discretionary spending. Notably, the scatter plot shows a denser cluster of data points at the lower end of the income spectrum, spreading out as income rises, which is indicative of increased variability in spending behaviors among higher earners. The Loess curve adeptly captures overall trend, highlighting the stronger influence of income on spending, especially in the mid to upper income ranges.

- **Are there unique spending behaviors specific to certain age groups? If so, what are these behaviors and what could be driving them?**

Total Amount Spent VS. Income, Faceted By Age Group



The faceted Degree 2 Loess plot illustrates distinct spending habits across various age groups when compared with income. I segmented the age into five different groups ("35 and Below", "36-45", "46-55", "56-65", "66-75"). The plots use a degree 2 Loess curve with a span of 1 to show the trend within each age group.

Trend breakdown:

35 and Below (Light Blue): This group appears to have a moderate upward trend. The Loess curve suggests that as income increases, the total amount spent also increases, but not as steeply as in some other age groups.

36-45 (Orange): This group shows a much steeper increase in spending as income increases, indicated by the steep Loess curve. It suggests that individuals in this age group tend to spend more as they earn more, compared to the 35 and below age group.

46-55 (Dark Blue): The trend here is similar to the 35 and below group but with a bit more variance. The spending increases with income but not as aggressively as in the 36-45 age groups.

56-65 (Green): This group has a curve similar to the 35 and below age group but with a lower starting point of spending for the same income levels. This could indicate a tendency towards saving or lower spending despite increases in income.

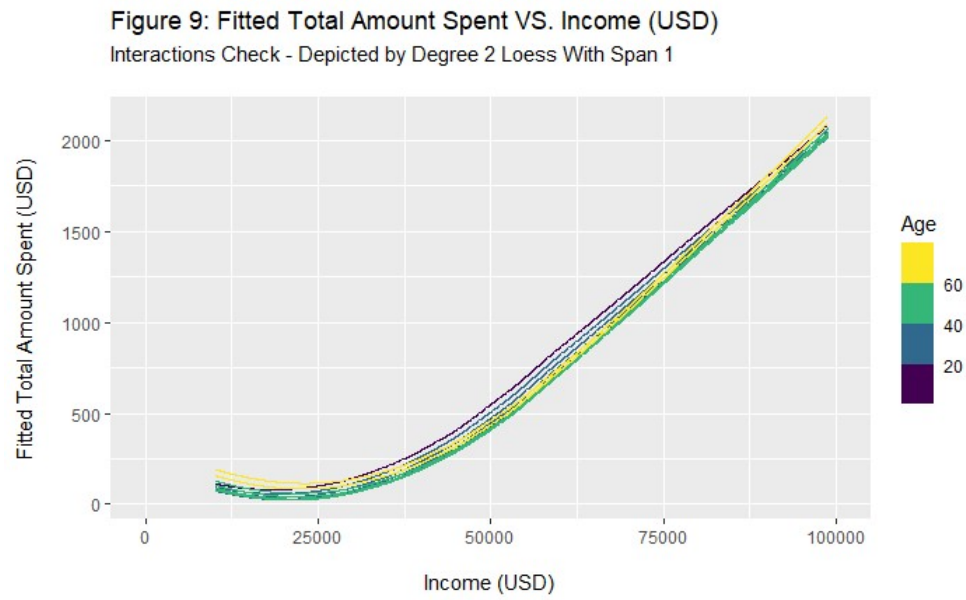
66-75 (Purple): The oldest group has the most distinctive pattern. While there is an upward trend in spending with increased income, it is not as steep as in the younger age groups, and the spending levels off at higher income levels, which might suggest a cap on spending regardless of additional income, possibly due to fixed retirement plans or lower propensity to spend. However, there are very few data points to generalize any spending trend.

In summary, the spending patterns are almost consistent across all age groups. However, the 36-45 age group demonstrates a unique behavior with a steep increase in spending as income increases and the 66-75 group also shows a distinct pattern where spending levels off at higher incomes, possibly due to retirement budgeting or reduced spending habits.

The analysis suggests that spending behaviors are not only influenced by income but also by the life stage associated with each age group. Understanding these distinctions is pivotal for businesses to tailor their products and services to meet the nuanced needs of each demographic segment.

Considering this, fitting a degree 2 loess curve with a span of 1 could be further explored.

Checking Interactions:



Do we need an interaction between Income and Age?

The plot illustrates the relationship between income and spending across different age groups, captured by a color gradient that represents age from young (purple) to old (yellow).The Loess curves, following a degree 2 model with a span of 1, indicate a pattern in spending behavior relative to income.

Initially, there's a considerable spread between the age groups at lower income levels. However, as income increases, the spread between the age groups' spending appears to tighten, suggesting that the differences in spending by age become less pronounced at higher income levels. In the mid-income range, the spread between the curves widens slightly, before narrowing again at the highest incomes shown.

The observed variation in this difference suggests the presence of interactions, indicating that the relationship between income and spending is indeed affected by age.

Let's now fit a model to examine the interaction between Spending vs. Income and Age.

```
customer.lo <- loess(Total_Amount_Spent ~ Income * Age, data = customer.data, degree = 2,
                     span = 1, normalize = FALSE)
```

Residual Plots: Lack-of-trend check

1) Residuals Against Age, Faceted By Income (Least Square Loess Model), *Refer to Appendix - Figure-17*

The residual plot depicts the relationship between residuals and age, faceted by income. The presented data illustrates that the residuals are distributed randomly about the x-axis for income ranging from 40000 to 100000 USD. This pattern suggests that the loess model (degree 2 and span 1) may be a most suitable fit for elucidating the relationship between spending from income and age. However, it is noticeable that the residuals are not scattered randomly about the x axis for the initial income group (upto 40000 USD). This discrepancy could be attributed to the limited number of data points within this income range.

2) Residuals Against Income, Faceted By Age (Least Square Loess Model), *Refer to Appendix - Figure-18*

The residual plot represents the relationship between residuals and income, faceted by the age. The plotted data illustrates that, for the age group from "35 and Below" to "66-75", the residuals are randomly distributed around the x-axis. This pattern suggests that the loess model (degree 2 and span 1) could be the most suitable fit for understanding the relationship between spending from income and age.

Taking into account the presence of outliers, I attempted to fit the robust loess curve. The plot depicting **robust fit residuals against the age, faceted by income** (*Refer to Appendix - Figure-19*) and **robust fit residuals against the income, faceted by age** (*Refer to Appendix - Figure-20*), closely resembles the plot of least square residuals for the same parameters.

After experimenting with different spans for our degree 2 loess model with an interaction, I settled on a span of 1. This choice provided a better fit for a model with an interaction, and the residual plots also indicated that the residuals are distributed randomly about the x-axis. This pattern suggests that the loess model (degree 2 and span 1) with an interaction could be the most suitable fit for elucidating the relationship between spending from income and age.

Therefore, I concluded to finalize the least square degree 2 Loess Model with a Span of 1. This choice represents the most suitable approximate fit for the dataset, effectively explaining the relationship between spending from income and age.

Goodness-of-fit: R-Squared

An R-squared value of 0.7394 for the model suggests that approximately 73.94% of the variation in the total amount spent can be explained by income across the different age groups. This high R-squared value indicates that the model has a relatively strong fit to the data, implying that income is a significant predictor of spending. While this model appears to capture a substantial portion of the spending behavior, there is still around 26.06% of the variation that is not explained by income alone. This unexplained variance may be due to other factors not included in the model, such as individual or cultural differences in spending habits, savings rates, or other economic factors. It might be insightful to consider additional variables that could affect spending to improve the model's explanatory power further.

Research Question 2

- **How does marital status influence spending relative to income?**
- **Is the spending-income relationship for different marital statuses best explained by a straightforward linear model, or does it require a more complex model to capture nuances?**

Relationship between Total Amount Spent and Income
(Refer to Figure-6)

The depicted plot illustrates the relationship between Total Amount Spent and Income, analyzed using three different models. The linear model is represented by the dark blue line, a loess curve of degree 2 with a span of 1 is depicted in light blue, and the orange curve represents a gam curve.

Upon careful examination of the data point distribution, it becomes apparent that the linear model is unsuitable choices, as they fail to capture the evident non-linear trend in the data.

The scatter plot, featuring data points along with various fit lines, reveals that the light blue curve (loess with a span of 1 and degree 2) strikes a good balance. However, the orange curve (gam) adeptly captures the overall trend while accommodating local variations when dealing with categorical variable like marital status.

Therefore, we observe that the **GAM model (i.e. Curved Function)** is good enough to define the relationship between Total Amount Spent and Income relative to Marital Status.

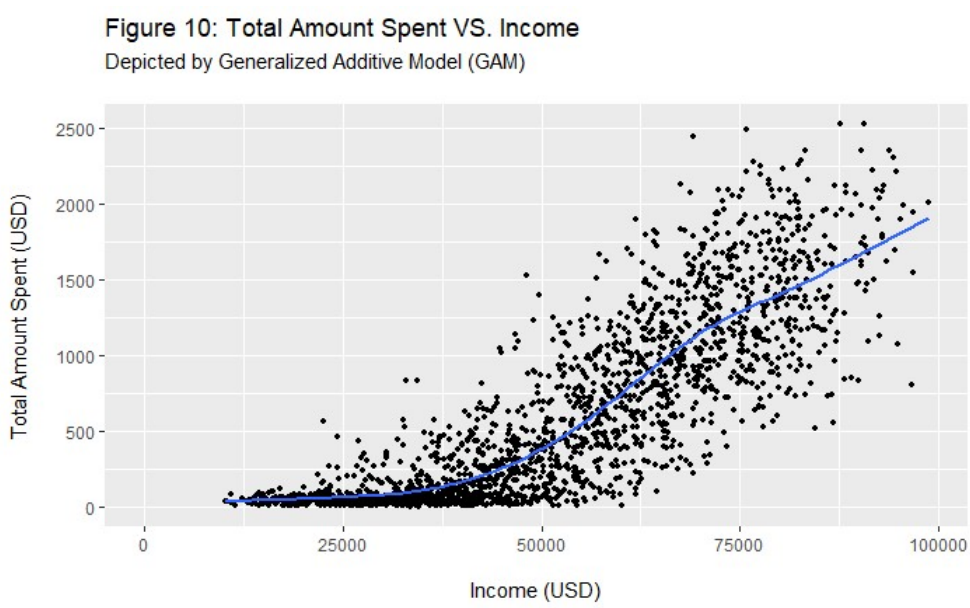


Figure 10 illustrates the relationship between income and total amount spent using a Generalized Additive Model (GAM), which allows for the estimation of a non-linear association. The scatter plot presents individual data points representing customers, with the blue line showing the smoothed trend as estimated by the GAM. Interpreting the GAM curve, we see a gradual increase in spending with income up to around $50,000, after which the spending accelerates more sharply, indicating that higher-income customers tend to spend significantly more. This trend flattens out somewhat as income approaches $100,000, suggesting an increase in income has a less pronounced impact on spending.

- **Are there identifiable patterns in spending behaviors among different marital statuses, and what might explain any variations?**

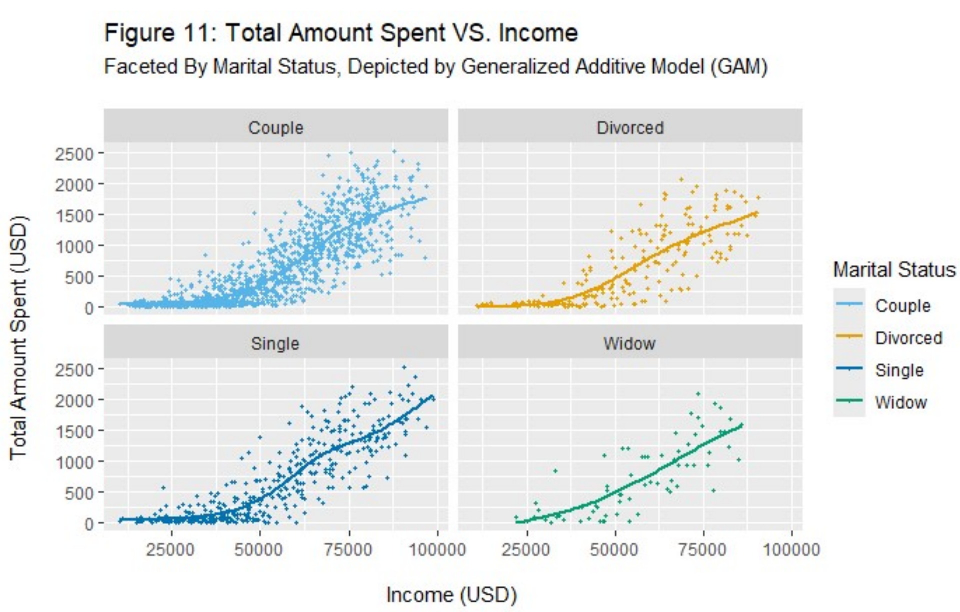


Figure 11, segmented by marital status and modeled using Generalized Additive Models (GAM), exhibits similar spending trends in relation to income among different marital categories except for the window category maybe due to less number of data points. However, the rate at which spending increases with income is different for each category.

There is a positive relationship between Income and Spending across all marital statuses.

Trend Breakdown by Marital Status:

Widows and **singles** seem to have a steeper slope, indicating a higher increase in spending with income compared to couples and divorced individuals.

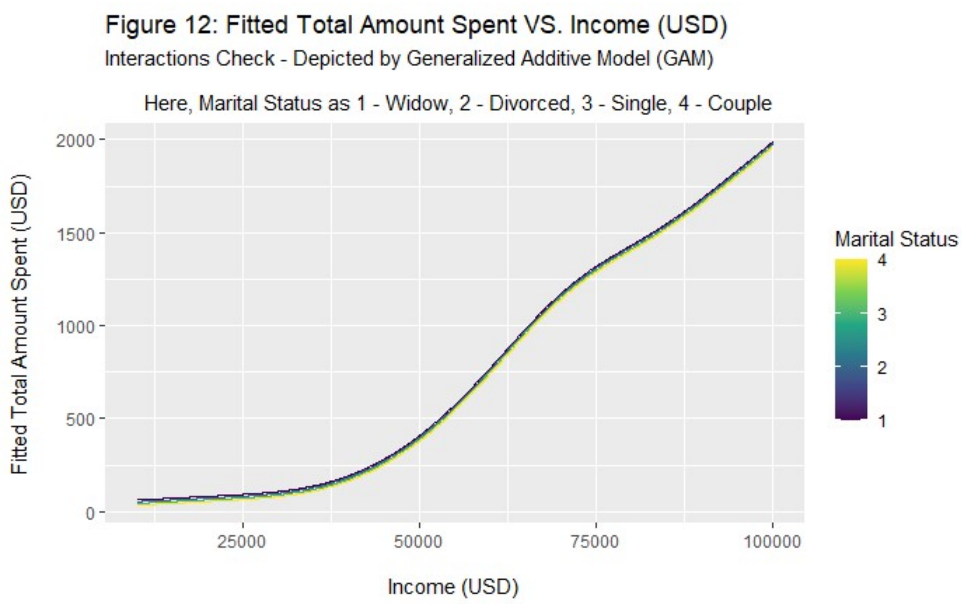
The trend for **couples** shows a consistent gradual increase in spending with income. However, the spending curve flattens as income approaches the higher end, suggesting a potential saturation point beyond which additional income does not translate into proportionally higher spending.

For those **divorced**, the trend is similar to couples. However, it is less steep, with spending increasing less rapidly with income when compared with couples. This group's behavior suggests a lifestyle changes post-divorce.

These variations among marital statuses reflect how spending behavior could be influenced by the economic and social factors pertinent to each group. These insights are invaluable for businesses seeking to customize their marketing and sales strategies to align with the fiscal behaviors of their diverse customer base.

Considering this, fitting a gam curve could be further explored.

Checking Interactions:



Do you need an interaction between Income and Marital Status?

We note that the curves for different marital statuses closely track each other and virtually overlap, suggesting minimal differentiation in spending behavior related to marital status at varying income levels. The lack of divergence between the lines indicates that the influence of income on spending is consistent regardless of marital status.

Therefore, the analysis does not support the necessity for an interaction term between income and marital status in the model. The spending trends appear uniform across different marital statuses, implying that marital status does not significantly modify the relationship between income and total spending.

Let's now fit a model to examine the relationship between income and total spending.

```
customer.gam <- gam(Total_Amount_Spent ~ s(Income), data = customer.data, method = "REML")
```

Residual Plots:

1) Lack-of-trend check: Residuals Against Income (Generalized Additive Model (GAM)), *Refer to Appendix - Figure-21*

The residual plot depicts the relationship between residuals and income. The presented data illustrates that the residuals are distributed randomly about the x-axis for income ranging from 40000 to 100000 USD. This pattern suggests that the gam model may be a most suitable fit for elucidating the relationship between spending from income. However, it is noticeable that the residuals are not scattered randomly about the x axis for the initial income group (upto 40000 USD). This discrepancy could be attributed to the limited number of data points within this income range.

2) No-interaction check: Residuals Against Income, Faceted by Marital Status (Generalized Additive Model (GAM)),
Refer to Appendix - Figure-22

This scatter plot represents the relationship between residuals and income, faceted by the marital status. The plotted data illustrates that, for the marital status widow, divorced, single and couple, the residuals are randomly distributed around the x-axis. This pattern suggests that the gam model could be the most suitable fit for understanding the relationship between spending and income categorized by marital status.

Therefore, I concluded to finalize the GAM Model. This choice represents the most suitable approximate fit for the dataset, effectively explaining the relationship between spending and income.

Goodness-of-fit: R-Squared

An R-squared value of 0.7504 for the model suggests that approximately 75.04% of the variation in the total amount spent can be explained by income.

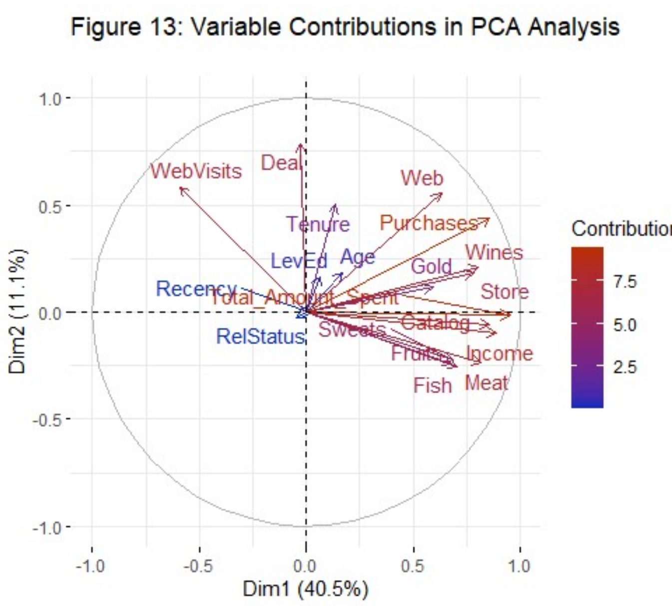
This high R-squared value indicates that the model has a relatively strong fit to the data, implying that income is a significant predictor of spending. While this model appears to capture a substantial portion of the spending behavior, there is still around 24.96% of the variation that is not explained by income alone. This unexplained variance may be due to other factors not included in the model, such as individual or cultural differences in spending habits, savings rates, or other economic factors. It might be insightful to consider additional variables that could affect spending to improve the model's explanatory power further.

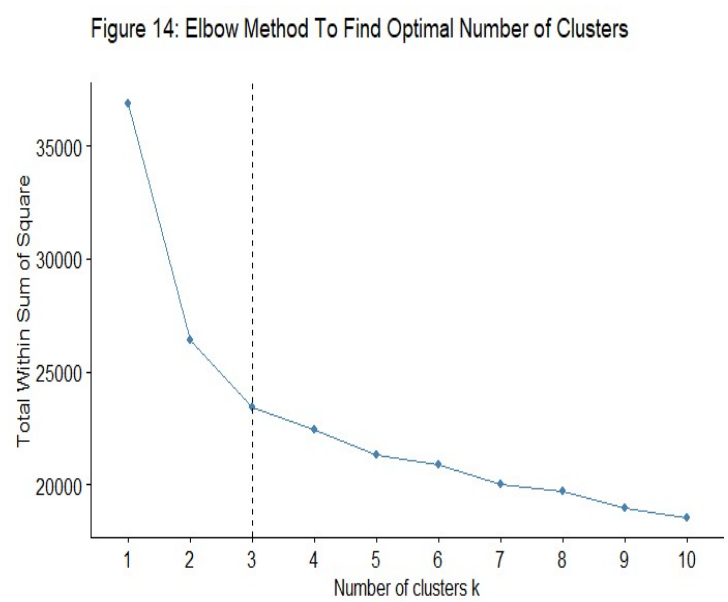
Research Question 3

- **What distinct segments of customers can be identified based on their purchasing behaviors, demographics, and interactions with marketing campaigns?**

In our pursuit of extracting meaningful customer segments, we meticulously prepared the dataset for Principal Component Analysis (PCA). This initial phase involved the careful exclusion of missing values, which could potentially skew our results. We also addressed outliers that represented extreme cases, not characteristic of our general customer base. Additionally, we streamlined the dataset by omitting non-quantitative variables and those features deemed less pertinent for capturing the essence of customer behaviors.

With the data thus curated, we execute Principal Component Analysis. To illustrate the impact of each variable on the principal components, we use a factor map that employs a color gradient to clearly demonstrate the magnitude of contributions, enhancing our understanding of the data's underlying structure. (*Refer to Figure-13)*





Integrating Principal Components for Enhanced Customer Insight: *The Significance of Both Spending Patterns and Engagement Behaviors Captured by PC1 and PC2 in Customer Segmentation*

In PCA, each principal component captures different aspects of variance in the dataset. PC1 typically captures the most significant variance, and PC2 captures the second most, but importantly, it captures variance that is orthogonal to PC1, meaning it captures a different type of variation within the data.

From the plots and summaries, we can infer the following about PC1 and PC2:

PC1 (Dim1) is largely associated with variables that represent overall spending and purchasing behavior like "Total_Amount_Spent", "Income", "Catalog", "Purchases" and "Meat". This suggests that PC1 may represent a "spending power" or "purchasing volume" dimension, differentiating customers based on how much they spend and how often they make purchases.

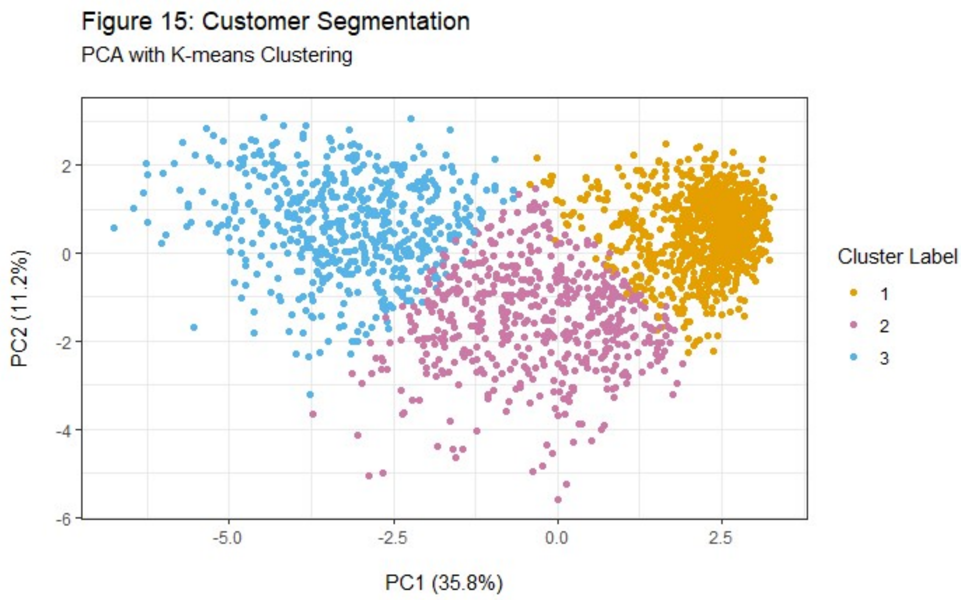
PC2 (Dim2) on the other hand, shows a strong relationship with variables such as "Deal", "WebVisits", "Web", "Tenure" and "Purchases." This suggests that PC2 may capture behaviors related to customers' engagement with marketing campaigns, their tenure or loyalty, and their interaction with different sales channels (like the web, store, catalog and deals). The negative correlations with "Sweets", "Meat" and "Fruits" could suggest that PC2 also contrasts purchasing patterns in terms of product categories.

The combination of PC1 and PC2 provides a more comprehensive view of customer behavior. PC1 might segment customers based on how much they spend overall, while PC2 could differentiate them based on how they respond to marketing, customer engagement (like web visits and deal usage), how they shop (online vs. offline), and what they buy. This multifaceted insight is crucial for businesses to tailor their marketing strategies, product offerings, and customer engagement approaches effectively. For example, focusing on upselling strategies for high-spenders or improving online user experience to convert web visits to purchases.

In the preliminary stages of our analysis, we recognized the importance of minimizing multicollinearity to preserve the integrity of the Principal Component Analysis (PCA) and subsequent clustering. To this end, we conducted a rigorous correlation analysis with a focus on detecting and eliminating highly correlated features. We adopted a threshold of 0.75, allowing us to discern and remove variables that were closely correlated, thus minimizing the risk of multicollinearity in our model.

Post-correlation analysis, the features that constituted our refined dataset included: **'Income', 'Recency', 'Tenure', 'Age', 'Wines', 'Fruits', 'Meat', 'Fish', 'Sweets', 'Gold', 'RelStatus', 'LevEd', 'WebVisits', 'Web', 'Deal', 'Catalog', and 'Store'**. This curated list of variables was poised to deliver a more authentic and distinct segmentation of our customer base. Upon streamlining our feature set, we proceeded with the normalization of the data. Moving forward with our chosen set of variables, we applied the K-means clustering algorithm to distill our customer data into meaningful groups. To ascertain the most effective number of clusters, we employed the **Elbow Method** - a technique that plots the within-cluster sum of squares (WSS) against the number of clusters. Upon careful evaluation, the 'elbow' was discerned at the point where **three clusters** were considered, leading us to select **K = 3 for our K-means algorithm**. This decision was visually substantiated by the clear inflection point on the plot *(Refer to Figure-14),* affirming that three clusters were sufficient to capture the intrinsic segments within our customer base without overcomplicating the model.

Customer Segmentation - *Combining PCA and K-means Clustering*



Interpretation:

This scatter plot, derived from the application of K-means clustering on the principal component analysis (PCA) transformed data, reveals the segmentation of customers into three distinct clusters. Each dot represents a customer, positioned according to their scores on the first two principal components - PC1 and PC2, which explain 35.8% and 11.2% of the variance, respectively.

Cluster 1: Shown in Orange, forms a relatively tight grouping, suggesting high similarity within this segment. These customers are positioned towards higher values on PC1 and span across PC2. This suggests that Cluster 1's behavior is strongly aligned with the dominant variance in purchasing behaviors captured by PC1. The spread along PC2 indicates variability in the secondary behaviors or characteristics captured by this component.

Cluster 2: Shown in Purple, This group is centrally located around the lower to middle range of PC1 and spreads across the negative and positive values of PC2. This central positioning suggests that these customers exhibit average spending behaviors and have a mixed response to the behavioral aspects captured by PC2.

Cluster 3: Shown in Blue, this cluster is distinguished by higher PC2 scores. These customers are predominantly found towards the lower end of PC1. The distribution along PC2 suggests diverse behaviors in aspects not captured by the primary spending variance, possibly indicating this group's varied interactions with marketing campaigns or engagement with different sales channels.

The clusters exhibit some degree of overlap, especially between clusters 1 and 2, suggesting that there might be a transition zone where customer behaviors and spending are not sharply distinct. However, each cluster is sufficiently separated to suggest that they represent different customer behaviors and preferences.

- **What unique characteristics define each identified customer segment?**

Our comprehensive analysis has identified three distinct customer segments, each with unique purchasing behaviors, demographic profiles, and interactions with marketing campaigns. The characteristics of each segment are as follows:

Cluster 1: Emerging Economicals

- **Income:** This segment represents customers with the lowest income bracket, suggesting limited spending capacity.
- **Age:** They tend to be younger, with a median age of approximately 41, and are the most recent members of the company's customer base.
- **Digital Engagement:** They visit the company's website more often than other segments, yet their spending via web and overall is the lowest, indicating a more cautious or price-conscious shopping behavior.
- **Segmentation Label:** Due to their high digital engagement but low spending, customers in Cluster 1 can be identified as "Emerging Economicals" or "Conservative Shoppers".

Cluster 2: Established Balancers

- **Income:** Customers in this cluster have average income levels, indicating a middle-ground spending potential.
- **Age:** They tend to have a median age of approximately 50.
- **Tenure:** They have been associated with the company for the longest duration, indicating loyalty and consistent patronage.
- **Shopping Habits:** Their spending is moderate in each product category, suggesting a balanced approach to purchases.
- **Engagement:** They frequently visit the website, potentially for browsing or engaging with content, and utilize various shopping modes, including web, store, and deals.
- **Segmentation Label:** Reflecting their loyalty and balanced shopping habits, Cluster 2 can be termed "Established Balancers" or "Balanced Value Seekers".

Cluster 3: High-Income Enthusiasts

- **Income:** This segment has the highest income levels among the three clusters.
- **Demographics:** With a median age of about 45, this group might consist of established professionals who prioritize quality over discounts.
- **Spending Behavior:** They exhibit the highest spending across all product categories, particularly on wines and meat, suggesting a preference for premium products.
- **Purchasing Channels:** Despite their high spending, they tend not to utilize deals, indicating a lower sensitivity to price or promotions. Despite their high income, their actual spending via web channels is lower compared to their potential, suggesting a behavior of browsing or researching online without necessarily leading to proportionate online purchases.
- **Segmentation Label:** Given their high spending irrespective of deals, customers in Cluster 3 can be classified as "High-Income Enthusiasts" or "Affluent Explorers".

Strategic Implications

By categorizing customers into these segments - High-Income Enthusiasts, Established Balancers, and Emerging Economicals - companies can fine-tune their marketing strategies and customer engagement approaches. For High-Income Enthusiasts, companies can focus on exclusive offers and premium product lines. For Established Balancers, personalized services, loyalty programs, and cross-channel engagement could drive retention. Lastly, for Emerging Economicals, enhancing the online user experience and offering value deals could convert website visits into purchases. Understanding these customer segments allows businesses to make data-informed decisions to enhance customer satisfaction, foster loyalty, and drive growth.

Conclusions

Spending vs. Income by Age: The research indicated that spending behavior does indeed vary across different age groups. The analysis illustrates a general increase in spending with rising household income across all age groups. Notably, younger customers (35 and below) exhibit more conservative spending regardless of income, suggesting that other factors may influence their purchasing decisions. The customers under 36-45 age groups exhibited a pronounced increase in spending as income rose, whereas the 66-75 age groups showed a levelling off in spending at higher incomes, potentially due to fixed retirement income or reduced spending habits. This suggests that life stage considerations are crucial in understanding consumer spending behavior, with income being a significant predictor of spending across age groups.

Spending vs. Income by Marital Status: The analysis showed that spending relative to income was consistent across different marital statuses, suggesting that marital status does not significantly modify the relationship between income and spending. This conclusion is drawn from the observation that the lines representing different marital statuses in the gam model closely tracked each other, indicating uniform spending trends irrespective of marital status.

Customer Segmentation: The PCA analysis revealed key variables that define customer segments. Variables such as Total Amount Spent, Income, Catalog, Purchases, and Meat were significant in explaining customer behavior variability. The results indicated that financial capacity i.e. income and the types of products purchased are primary indicators of customer expenditure behavior, which can guide businesses in tailoring their products and services more effectively to meet the needs of each customer segment. Additionally, clustering methods have effectively segmented the customer base into three distinct groups i.e. High-Income Enthusiasts, Established Balancers, and Emerging Economicals. These segments demonstrate unique purchasing patterns and demographic features which can be leveraged for more targeted marketing strategies.

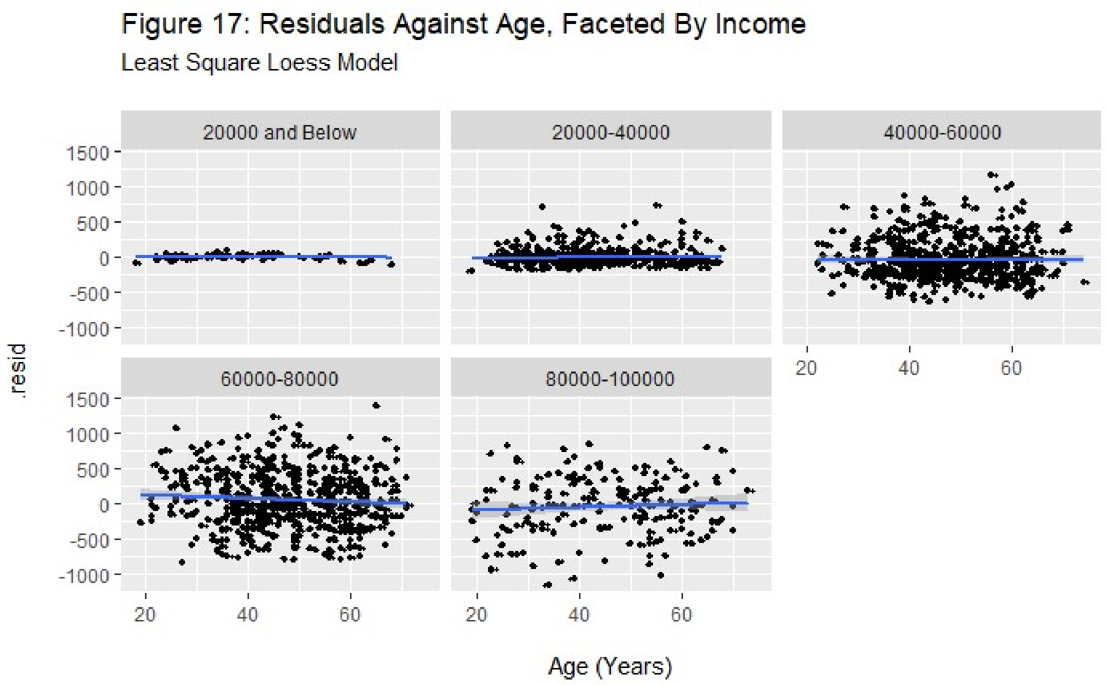
Limitations

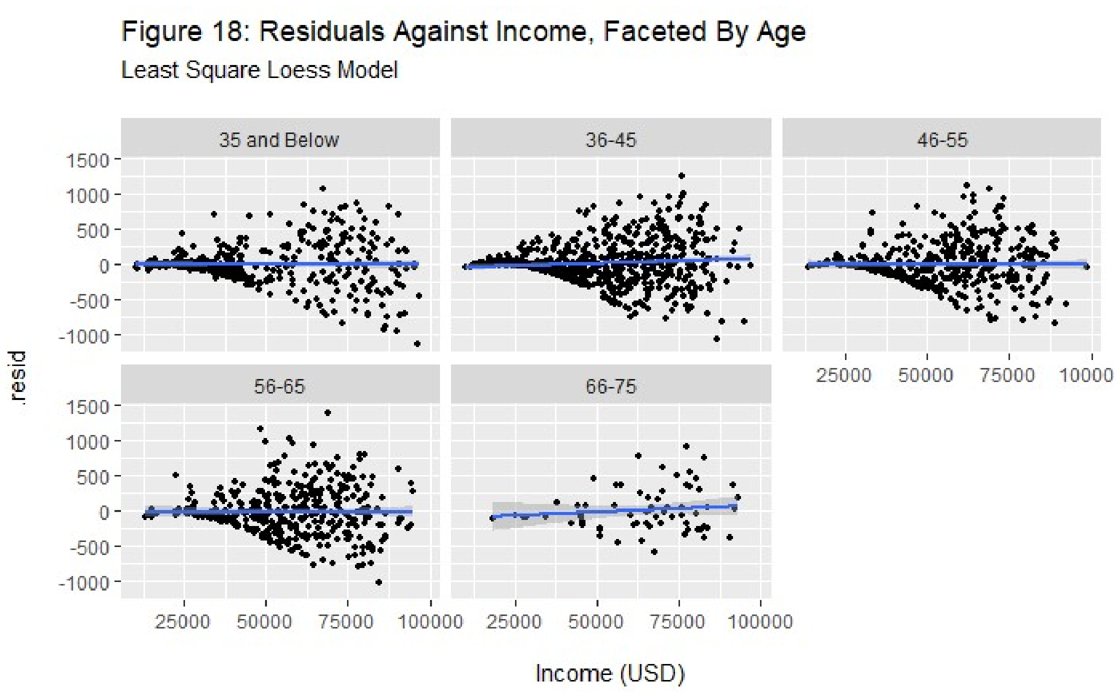
The analysis is limited to the variables available in the dataset. There might be external factors not included in the model, such as cultural differences, economic factors, or psychological factors that influence spending habits. The model explains a significant proportion of variance in spending, but a non-negligible amount of variance remains unexplained, which could be important in understanding the complete picture of consumer behavior.

Future Scope

Future studies could incorporate additional variables that may influence spending, such as educational level, number of dependents, or more nuanced measures of engagement with marketing campaigns. Longitudinal data could be used to examine changes in spending behavior over time and the impact of life events on consumer spending. Exploring machine learning algorithms like random forests or neural networks could reveal non-linear patterns and interactions not detected by the current models. Further research could explore the psychological drivers of spending behavior to provide a more holistic view of consumer decision-making processes.

Appendix: Compilation of figures employed in the report





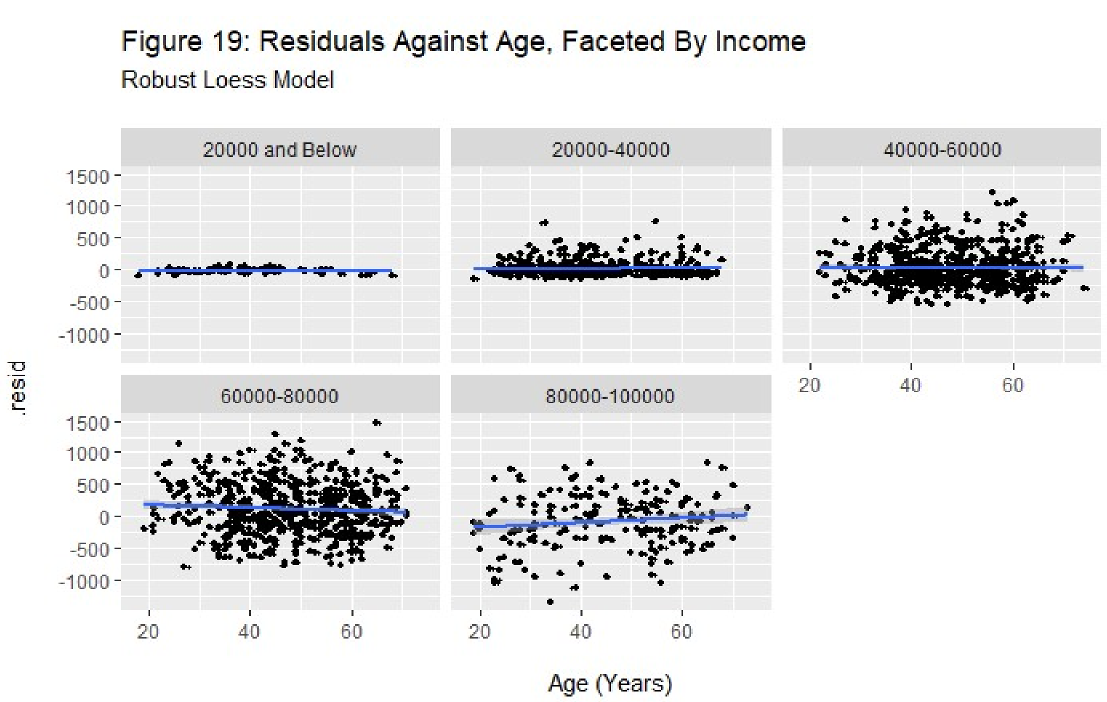


Figure 20: Residuals Against Income, Faceted By Age

Robust Loess Model

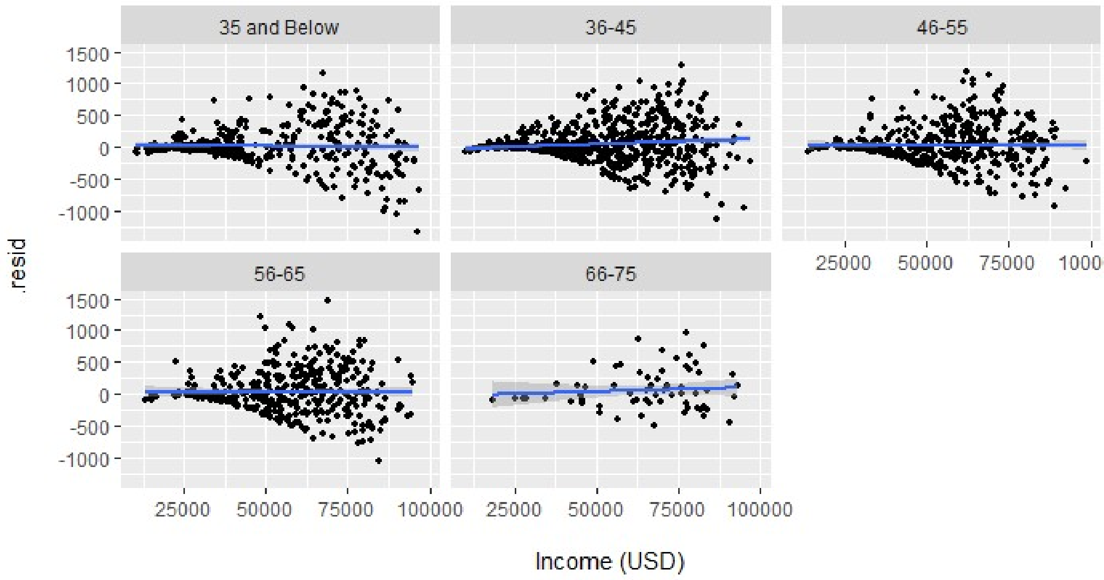


Figure 21: Residuals Against Income

Generalized Additive Model (GAM)

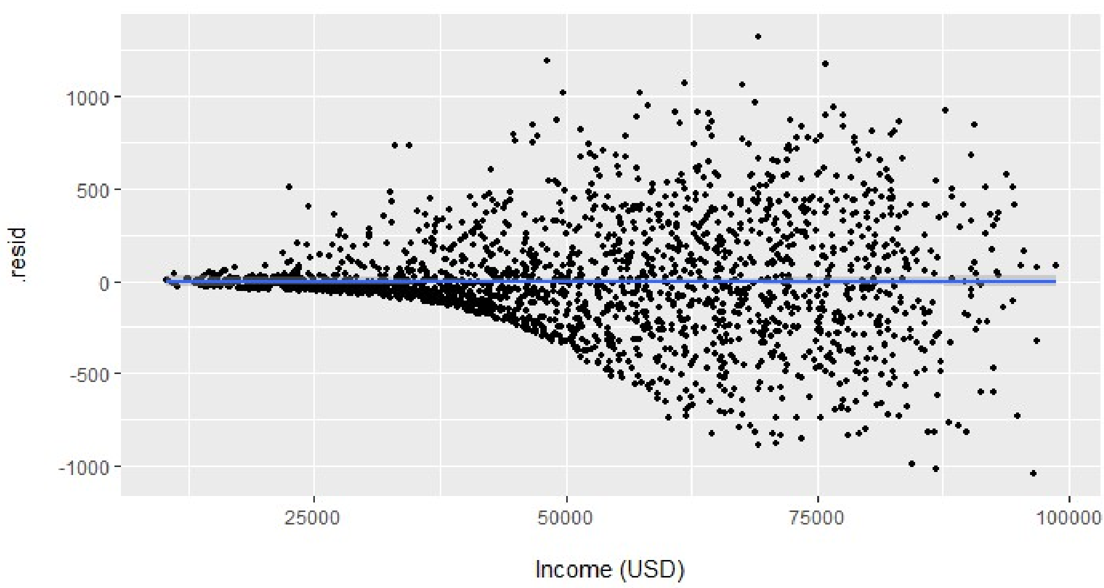


Figure 22: Residuals Against Income, Faceted By Marital Status

Generalized Additive Model (GAM)